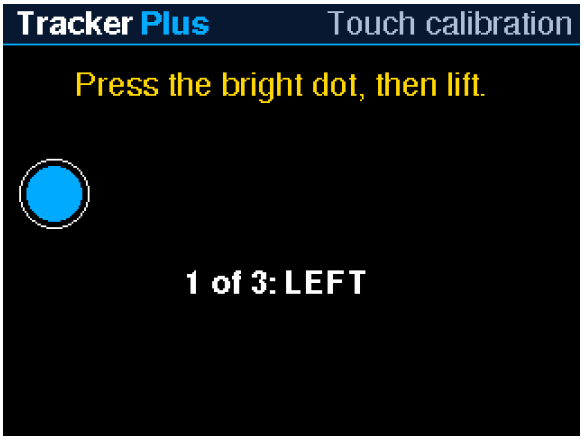
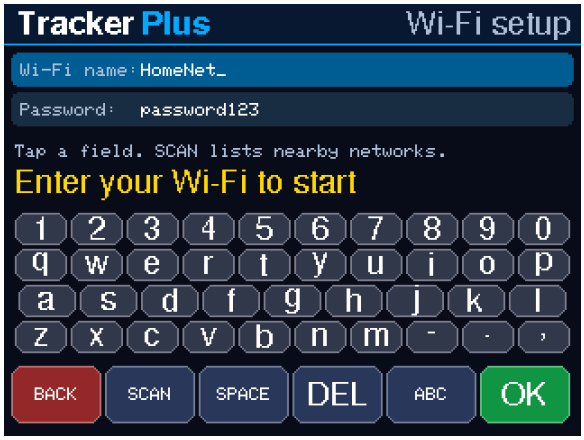
Tracker Plus

How-to guide — 6 apps in 1: Sports, Soccer, Stock Ticker, Aircraft, Ships and World Events on one CYD2usb (ESP32-2432S028C) — version 1.3

1. First start



Touch calibration: press each dot, then lift

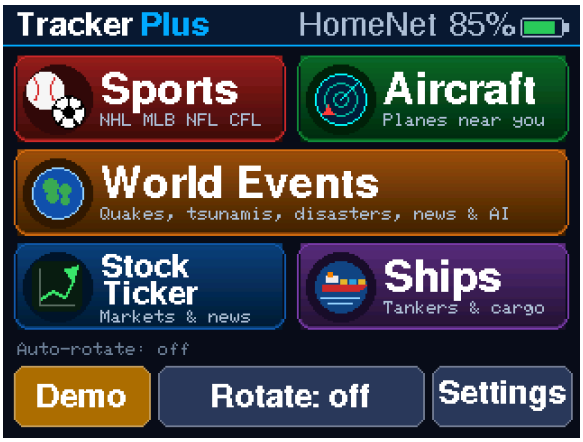


Wi-Fi setup (hold the screen 3 s to calibrate again)

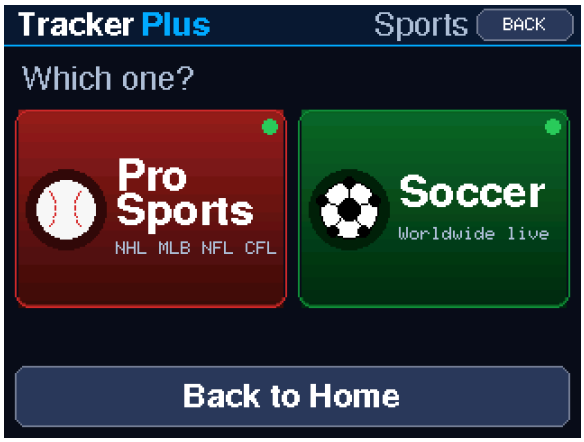
• A fresh unit calibrates the touch screen first, then shows the Wi-Fi screen. Type your network or tap **SCAN** to pick it from a list, type the password, then **OK**.

• Calibration and Wi-Fi are saved once and shared by all six apps. All setup lives in one place: **Home > Settings**. The apps no longer have Wi-Fi or calibration screens of their own; the **Settings** button in each app's menu takes you straight to Home > Settings.

2. Home menu



Home: tap a tile to open that app



Sports: pick Pro Sports or Soccer

• Five tiles: **Sports** and **Aircraft** at the top, **World Events** across the middle, **Stock Ticker** and **Ships** at the bottom. Sports then asks **Pro Sports** (NHL, MLB, NFL, CFL) or **Soccer** (leagues and cups worldwide). **Back to Home** returns.

• Bottom row: **Demo**, **Rotate** and **Settings**.

• Top right: your Wi-Fi name and, on units with a battery, the battery level (for example 85%). While the charger is plugged in, a **lightning bolt** shows inside the battery (in the apps, the bolt shows in place of the %). **Units without a battery show no battery icon at all — that is normal.**

• Only one app runs at a time. Switching restarts the board quietly (about 3–5 s) so the app you open always gets all the memory.

• Please be patient when opening an app. It downloads fresh data each time, so it can take a few seconds (Pro Sports and Soccer up to about a minute the first time, while they load schedules and team logos). Stock Ticker shows its screen straight away and fills in as the data arrives.

• To come back Home: open the app's menu (tap **SET** at the top right, or the top right corner in Stock Ticker) and tap **Home**. Aircraft has a **Home** button at the bottom right.

• Switching the power off and on always starts at Home (or starts rotating if auto-rotate is on).

• Your teams, map place, time zone and so on are kept inside each app.

3. Pro Sports, Soccer and Stock Ticker

• **Pro Sports**: the first time, pick your teams (tap a league, tap teams; **up to 12**). Live scores, countdowns, schedules, results and scoreboards. Change teams any time: SET > **My teams**. A unit that was set up with more than 12 teams keeps its first 12.

• **Soccer**: pick a league, then your teams (**up to 12**). Live matches, next match, league table and scores from around the world. SET > **My teams** to change.

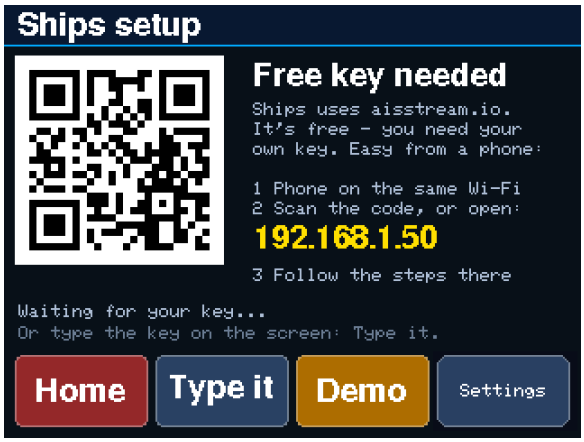
• **Stock Ticker**: clock, weather, stocks, crypto and news. Nothing to enter: location and time zone are found automatically from the internet connection. Its menu (top right corner): Units (Celsius/Fahrenheit), Refresh everything now, Settings, Home, Done.

• The FAQ for each app (scan the QR code under Settings > Help & info) explains every screen.

4. Aircraft and Ships



Aircraft: the radar on the left, the picked plane on the right



Ships: set up the free key with your phone

• **Aircraft**: the first time, it finds your location from the internet connection and shows it (for example Toronto, Ontario). Tap **OK**, or **Change** to type exact numbers. Change it later in Settings > **Radar location**.

• The radar shows every plane within 40 nautical miles, pointing the way it flies; the closest ones get their flight and height. The panel on the right shows the picked plane: flight, aircraft type, altitude, speed and distance. Tap a plane to pick it, or **Next** to step through them. It updates every 15 seconds. **SET** (top right) goes to Home > Settings.

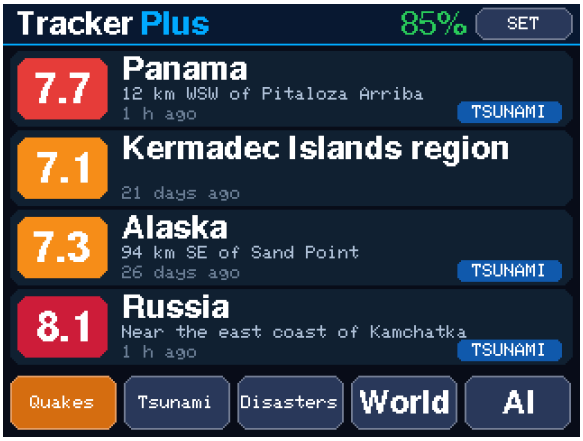
• Finding your latitude and longitude: in Google Maps on a phone, press and hold on your home until a red pin drops; the two numbers appear in the search bar (first latitude, then longitude). On a computer, right-click your home at maps.google.com. Keep the minus signs; two digits after the dot is plenty.

• **Ships** shows live tankers and cargo ships on a sea chart. It needs your own **free aisstream.io key** (the only thing you ever sign up for). The first time, Ships shows a QR code: with your phone on the same Wi-Fi, scan it (or type the address shown) and follow the steps there — sign in to aisstream.io, make a key, paste it. The key is saved on the unit. Or tap **Type it** to enter the key on the screen.

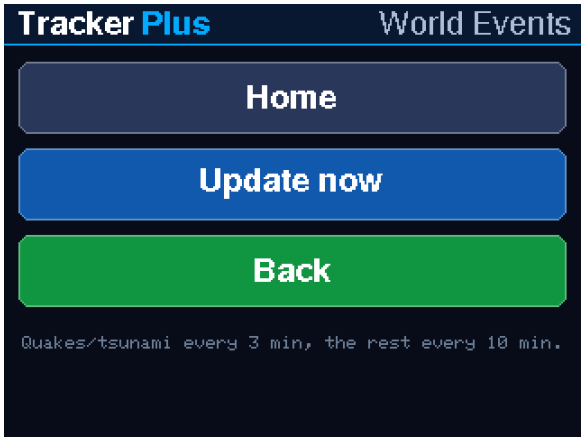
• Tap **Demo** on that screen to see Ships with sample ships, no key needed. SET > **AIS key** ends the demo and sets up the real key.

• In Ships: – and **+** at the top zoom the chart out and in. SET > **Find place** or **Hot spots** to pick a port or strait, SET > **AIS key** to change the key. New ships first appear as grey dots and turn red (tanker) or green (cargo) when their type arrives, which can take about 6 minutes.

5. World Events



World Events: five pages, tap a tab to change



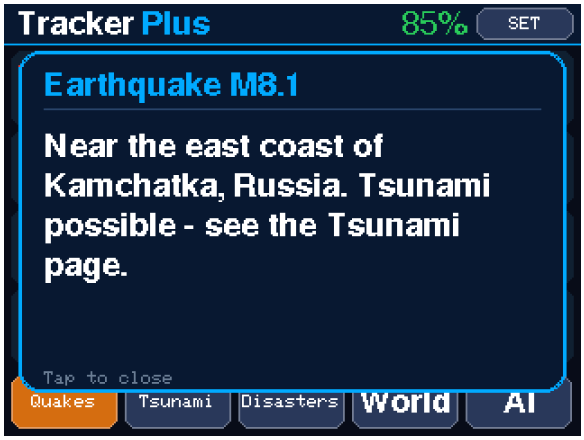
SET: Home, Update now, Back

• Five pages, picked with the tabs along the bottom: **Quakes** (magnitude 7 and up, last 30 days, from USGS), **Tsunami** (the Pacific and the US & Canada tsunami warning centres), **Disasters** (orange and red alerts from GDACS: storms, eruptions, floods, droughts), **World** (BBC World headlines) and **AI** (TechCrunch AI and MIT Technology Review, newest first).

• When a page has more than four items, tap its orange tab again to see the next few. Left untouched for a minute, it steps through the pages by itself (10 s each).



Tap a story: more detail and a QR code



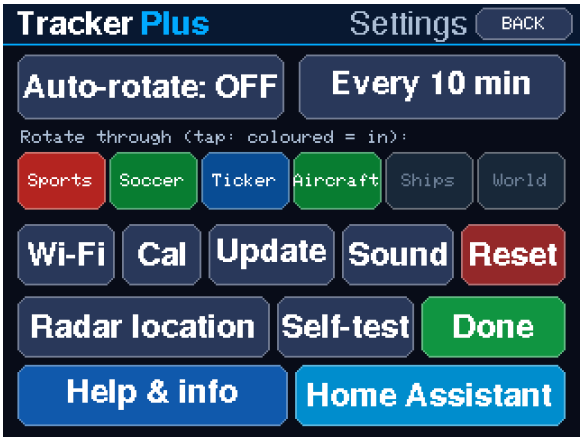
An alert for a major event

• **Tap any item** to open it: the full headline and summary (for quakes: place, depth, local time and the USGS alert level; for disasters: dates and severity; for tsunami messages: the bulletin text). Scan the **QR code** with a phone to read the full story. **BACK** returns.

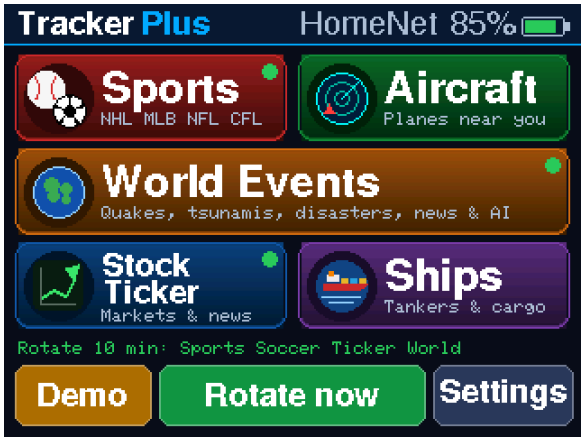
• Quakes and tsunami messages are checked every 3 minutes, the rest every 10 minutes. SET > **Update now** fetches everything straight away.

• **Alerts**: a card pops up with a sound (“World alerts”, see section 7) for a new magnitude 7+ quake in the last 6 hours, a tsunami warning, advisory, watch or threat in the last 6 hours, or a new red disaster alert. Tap the card to close it. Each alert shows once, even when auto-rotate restarts the app.

6. Settings and auto-rotate



Home > Settings



Auto-rotate on: green dots show the apps in the rotation

• **Auto-rotate ON/OFF**: when on, the board moves to the next chosen app after the chosen time.

• **Every N min**: tap to cycle through 2, 5, 10, 15 and 30 minutes.

• **Rotate through**: six buttons, one per app (Sports, Soccer, Ticker, Aircraft, Ships, World). Tap one to put it in the rotation (it lights up in its own colour) or take it out (dim). A green dot on the Home tile shows it is in the rotation.

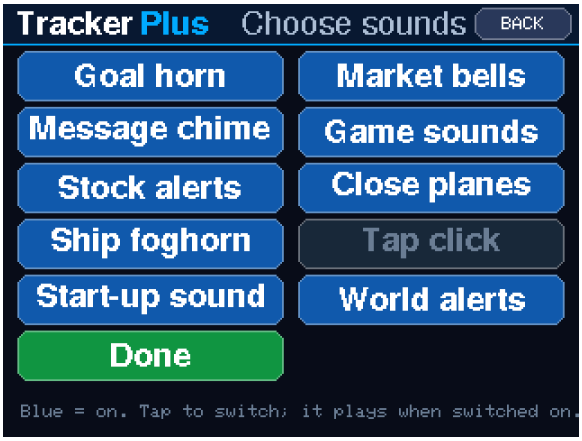
• Rotation waits while you are in a menu, while a goal or score flash is showing, and while one of your Pro Sports or Soccer teams is playing live (as long as its scores have updated in the last 10 minutes).

• Each switch reloads that app's data, so it needs a few moments to fill in. Choose 10 min or more to spend less time waiting and more time watching.

• With auto-rotate on, the board starts rotating at power-up. Home pauses it, and **Rotate now** on the Home screen restarts it.

• **Wi-Fi**, **Cal** (touch calibration), **Update**, **Sound**, **Reset**, **Radar location**, **Self-test**, **Help & info** and **Home Assistant** are here too.

7. Sound



Settings > Sound > Choose sounds

• Settings > **Sound**. Volume: **Off**, **Low**, **Medium** or **High** (the speaker plugs into the P4 connector).

• **Quiet: ON** keeps the unit silent at night (10 PM – 7 AM to start with). Tap the times next to it to choose other hours.

• **Choose sounds**: switch each sound on or off — goal horn, market bells, message chime (Home Assistant messages), game sounds, stock alerts, close planes, ship foghorn, tap click, start-up sound and **World alerts**. Blue means on; a sound plays when you switch it on. World alerts start switched on.

• **Test sound** plays every sound in turn. There are no sounds during the Demo.

8. Home Assistant (optional)

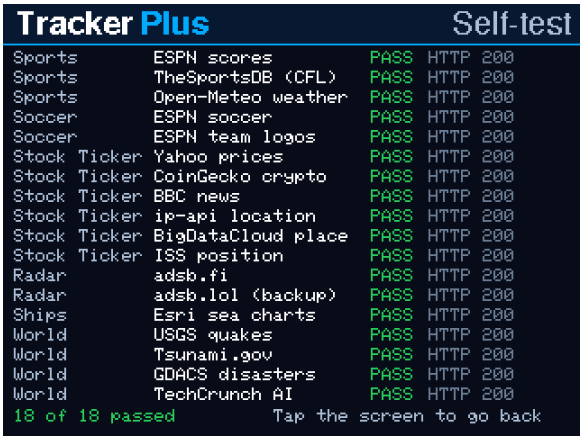
• For Home Assistant users. It needs the MQTT broker in Home Assistant (for example the Mosquitto add-on). Nothing happens until you set it up.

• Settings > **Home Assistant** shows a QR code and an address. With your phone on the same Wi-Fi, scan it (or type the address) and enter your broker's IP address, port (usually 1883), user name and password.

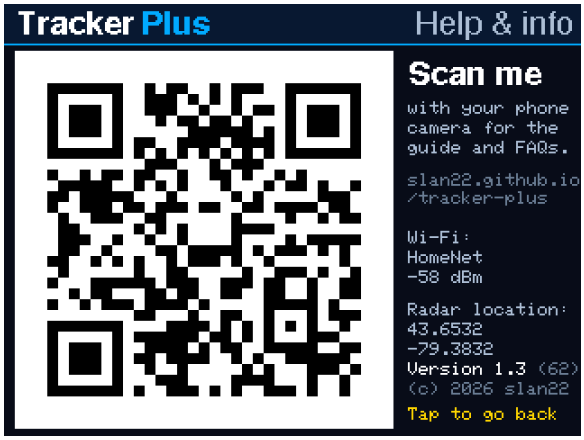
• Home Assistant finds the unit by itself: Settings > Devices & services > MQTT. From there you can pick the app showing (World Events included), switch the screen on or off, set brightness, turn auto-rotate on, start the Demo, go to the next app, and show a message on the screen (shown for 30 seconds; Notify works from automations).

• To stop it, tap **Turn off** on the Home Assistant screen.

9. Self-test and Help



Self-test: 18 checks, one per data source



Help & info: scan for this guide and the FAQs

• Settings > **Self-test** checks every internet source the six apps use, one by one (about a minute). PASS means that source answered; FAIL shows the reason. Tap the screen to go back.

• Aircraft has two sources (adsb.fi, with adsb.lol as backup), so one FAIL there still leaves Aircraft working. If something fails, check your Wi-Fi and try again later: the free data services are sometimes busy.

• Settings > **Help & info** shows a QR code. Scan it with a phone camera to open this guide and the FAQs at **slan22.github.io/tracker-plus**. The same screen shows the Wi-Fi name and signal, the Radar location and the version (1.3) with the build number (62).

10. Software updates over Wi-Fi

• Settings > **Update** checks for a newer build. Nothing is installed unless you tap **Install**. If a download fails, the board keeps running the build it had. Updates keep your Wi-Fi and settings.

• Updates come from the maker and are checked for authenticity before they are installed.

11. Wi-Fi

• The board remembers your router, so it usually joins in a couple of seconds.

• Once your Wi-Fi has worked, the board never gives up on it: if the router is slow or restarting (for example after a power cut) it shows *Still looking for [your network] - tap to change Wi-Fi* and keeps trying by itself.

• While it is connecting, tap the screen to open the Wi-Fi screen (new router or new password). Or go to Home > Settings > **Wi-Fi**.

• A network that has never worked: after 3 tries (about 30 s) the Wi-Fi screen comes back with the reason. Left alone, it tries the saved network again every minute.

• Aircraft stays on its screen when Wi-Fi drops and carries on when it is back.

12. Demo

• Tap **Demo** on Home. It shows every app in turn: Pro Sports and Soccer (starting with their team picker), Stock Ticker, Aircraft and Ships in the Strait of Hormuz with sample data, about 12 s each, then World Events with the real live news (3 s per page, about 20 s). Then it starts again.

• The demo keeps going until you open an app's menu and tap Home (or switch the power off and on). Wi-Fi is needed because the demo uses real logos, sea charts and news.

13. Troubleshooting

• Taps landing in the wrong place: Settings > Cal. Press the centre of each dot and hold it still for a moment. From the Wi-Fi screen, hold the screen for 3 s.

• An app looks empty or says Loading right after opening: give it a minute, it is downloading fresh data. If it still shows nothing, run Settings > Self-test to see which source is not answering.

• A World Events page says Loading for a while: if the first download fails it tries again on its next check (3 minutes for Quakes and Tsunami, 10 minutes for the rest), or tap SET > Update now.

• Ships shows no ships or only grey dots: give a new area 5–10 minutes, or try SET > Hot spots. A red status dot means the key is missing or rejected: SET > AIS key.

• No battery level at the top of Home: that unit has no battery fitted. This is normal. The battery level and the charging bolt are worked out from the battery voltage only, so the bolt can take a little while to appear or go away after the charger is plugged in or out.

• Pro Sports gets NHL, MLB and NFL games from ESPN (every game, preseason included). CFL comes from TheSportsDB.

• After a crash or power cut, the board comes back to Home (or starts rotating if auto-rotate is on).

Live data comes from free public services. Not affiliated with any league, team, exchange, airline, shipping company or news service.